SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	28/08/2026

Code of Conduct

I M.Mary Sharlin (192511330) *(Name / Reg No)* *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M.Mary Sharlin

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________	Signature: ______________	Signature: ______________
Date: ________________	Date: ________________	Date: ________________

Q1 **Minimum Spanning Tree - Kruskal's Algorithm** 20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

a. All zones must be connected

b. Total installation cost must be minimized

c. No redundant connections (no cycles) allowed

d. The network must remain fully connected

Your Answer

```
KRUSKAL(G)
1. MST ← empty set
2. Sort all edges of G in increasing order of weight
3. Create a separate set for each vertex
4. For each edge (u, v) in sorted edges:
    If FIND(u) ≠ FIND(v):
```

Save Answer Next →

Q2 **Graph - Topological Sort** 20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

a. Detect cycle before sorting

b. Use DFS approach

Your Answer

```
TOPOLOGICAL_SORT(G)
1. Mark all vertices as WHITE
2. Stack ← empty
3. For each vertex v in G:
    If v is WHITE:
        DFS(v)
```

Save Answer Next →

Q3 **Shortest Path Algorithm - MST & Dijkstra's Algorithm** 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
SMART_TRANSPORT(G, source)

1. MST ← KRUSKAL(G)
2. Build infrastructure using MST
3. Set distance[source] ← 0
4. Set all other distances ← ∞
```

Save Answer Next →

Q4 **Shortest Path Algorithm - Dijkstra's Algorithm** 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
KRUSKAL(G)

1. MST ← empty set
2. Sort all edges in increasing order of weight
3. Make each vertex a separate set
4. For each edge (u, v) in sorted order:
```

Save Answer Next →

Q5 **Minimum Spanning Tree - Prim's or Kruskal's Algorithm** 20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
SECOND_BEST_MST(G)

1. MST ← KRUSKAL(G)
2. Best ← ∞
3. For each edge e in MST:
    Remove e from MST
```

Save Answer